

Software Requirement Specification (SRS) Document

Group 8: Snakes and ladders

1. Brief Description of Project

This project is an implementation of the classic board game Snakes and Ladders, developed using the C programming language. It allows two or more players to take turns rolling a dice and moving across a 100-square board. The game logic includes:

- Snakes: Landing on the head of a snake moves the player down to its tail.
- Ladders: Landing at the bottom of a ladder moves the player up to its top.
- The first player to reach square 100 wins the game.

2. Purpose / Goal

The main goal of this project is Learn how to design and implement basic **game mechanics**, such as dice rolls, player movement, and win conditions

3. Usefulness / Benefit

- **For Users:** Ensures entertainment and fun along with a sense of nostalgia.
- **For Learners:** Strengthens programming fundamentals concepts like variables, loops, conditionals, arrays, and functions in C and develops Debugging and testing skills.

4. Hardware / Software Involved

Hardware Requirements:

- Standard PC/Laptop
- Minimum 2GB RAM
- Storage: Minimal space required

Software Requirements:

- Programming Language: C
- Compiler: GCC / Turbo C / Code::Blocks
- Operating System: Windows/Linux/macOS

5. Detailed Feature List

Core Gameplay Features

- Dice Roll Simulation: Generates random values between 1 and 6.
- Player Movement: Updates position based on dice roll.
- Board Representation: 100-cell board with snakes and ladders.

Game Rules & Logic

- Winning Condition: First player to reach 100 wins.
- Restricted Movement: Prevents exceeding position 100.
- Game Restart Option: Allows players to restart after a game ends.

6. Test / Demonstration Plan

- Validate Dice Rolls – Ensure output is always between 1 and 6.
- Test Player Movement – Verify correct position updates after each turn.
- Check Snakes & Ladders – Ensure players move up or down when landing on them.
- Verify Win Condition – Confirm game ends when a player reaches position 100.

7. Expected Interaction Interface and Sample Use Cases

Interaction Interface:

- the interaction is through **inputs** and **text display** in a terminal or command prompt window.

User interface features:

- Main Menu with numbered options (Start Game, Exit)
- Prompts to enter player names or count
- Turn-based dice roll
- Text output for dice result, player movement, ladder/snake encounter
- Final result display with winner announcement

Sample Use Cases:

1. Basic Gameplay Test

- **Scenario:** Player rolls a dice and moves forward.
- **Expected Outcome:** Player position updates correctly.

2. Snake Encounter

- **Scenario:** A player lands on a snake.
- **Expected Outcome:** Player position moves down as per snake rules.

3. Ladder Encounter

- **Scenario:** A player lands on a ladder.
- **Expected Outcome:** Player position moves up.

4. Winning Condition

- **Scenario:** Player reaches position 100.
- **Expected Outcome:** Game ends, declaring the player as the winner.

5. Invalid Moves

- **Scenario:** Player rolls a dice but exceeds position 100.
- **Expected Outcome:** Player remains at the previous position, waiting for an exact roll to win.

6. Randomized Dice Rolls

- **Scenario:** The dice is rolled multiple times.
- **Expected Outcome:** Outcomes are between 1 and 6, distributed fairly.